Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. [NEW] Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to Canvas.
8. Initial here to acknowledge that you did not use AI in completing this assignment, except where expressly allowed: ____

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks in your code chunks.

TIP: If your code fails to knit, check: * That no `install.packages()` or `View()` commands exist in your code. * That you are not displaying the entire contents of a large dataframe in your code.

Set up your R session

1. Load necessary packages (tidyverse, here), check your current working directory and import two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2018-08_raw.csv). Name these datasets "Neonics" and "Litter", respectively.

Be sure to: * Use the `here()` package in specifying the paths to your datasets * Include the appropriate subcommand to read in character based columns as factors

```
#Here I load libraries and CSV files

library(tidyverse)
library(here)
```

```
## Warning: package 'here' was built under R version 4.5.2
```

```
getwd()
```

```
## [1] "C:/Users/lexcl/OneDrive/Documents/EDE/EDE_Spring2026"
```

```
here()
```

```
## [1] "C:/Users/lexcl/OneDrive/Documents/EDE/EDE_Spring2026"
```

```
Neonics <- read.csv(file=here("./Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv"),stringsAsFactors =TRUE

Litter <- read.csv(file=here("./Data/Raw/NIWO_Litter/NEON_NIWO_Litter_massdata_2018-08_raw.csv"),string
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information. (AI is allowed here, but put answers in your own words.)

   Answer:

My guess is that the effects of pesticides observed on insects might serve as indicators of potential effects on other species, and/or the well-being of the ecosystem they inhabit. We might also be able to discern whether insects are developing resistance to the pesticides presumably designed to kill them.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information. (AI is allowed here, but put answers in your own words.)

   Answer:

Another guess here. If looking at the food people eat can help us understand a human's general health, examining litter and woody debris might help us gain insights into the health of the forest ground/soil. Such debris is also consumed by many species, including termites and other insects with an insatiable hankering for wood.

Maybe I am speculating now about how these two data sites are related, but it is many of these debris-eating species that are targeted by insecticides, potentially disrupting natural processes of decomposition/the nutrient cycle.

4. How is litter and woody debris sampled as part of the NEON network? Read the NEON_Litterfall_UserGuide.pdf document to learn more. List three pieces of salient information about the sampling methods here:

   Answer: 1.Litter and fine woody debris samplig is taken only at sites with vegetation > 2m tall and only in tower plots(page 3.) 2.In sites with >50% cover of woody vegetation, placement of litter traps is random.In sites with <50% cover, litter traps are placed in areas beneath the qualifying vegetation measure. 3.Sample data is nested in the following structure: subplotID < plotID < siteID < domainID.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```r
dim(Neonics)
```

```
## [1] 4623   30
```

```r
#answer: dimensions are 4623    30
```

6. Using the `summary` function on the "Effect" column, determine the most common effects that are studied. [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```r
#
summary(Neonics$Effect)%>%sort()
```

```
##       Hormone(s)        Histology       Physiology          Cell(s)
##                1                5                7                9
##     Biochemistry     Accumulation     Intoxication    Immunological
##               11               12               12               16
##       Morphology           Growth        Enzyme(s)         Genetics
##               22               38               62               82
##        Avoidance      Development     Reproduction Feeding behavior
##              102              136              197              255
##         Behavior        Mortality       Population
##              360             1493             1803
```

Question: Which two effects stand out as the most studied? Can you guess why these effects might specifically be of interest? > Answer: Mortality and Population stand out as the most studied presumably because they are the most closely associated with a species' survival.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name).[TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```r
summary(Neonics$Species.Group, maxsum =6)
```

```
##                                                Insects/Spiders
##                                                           3569
##                              Insects/Spiders; Standard Test Species
##                                                             27
## Insects/Spiders; Standard Test Species; U.S. Invasive Species
##                                                            667
##                              Insects/Spiders; U.S. Invasive Species
##                                                            360
```

```r
summary(Neonics$Species.Common.Name, maxsum =6)
```

```
##             Honey Bee        Parasitic Wasp Buff Tailed Bumblebee
##                   667                   285                   183
##   Carniolan Honey Bee            Bumble Bee               (Other)
##                   152                   140                  3196
```

```r
summary(Neonics$Test.Location, maxsum =6)
```

```
##     Field artificial        Field natural Field undeterminable
##                   96                 1663                    4
##                  Lab
##                 2860
```

Question: What do these species have in common? Why might they be of interest over other insects? > Answer: With the exception of the (other) group, they are all species of bees.Their role as pollinators makes them important.

8. The `Conc.1..Author` column, which lists the concentration of the neonicitoid dose, should include numeric values. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```r
class("Conc.1..Author")
```

```
## [1] "character"
```

```r
View(Neonics)
```

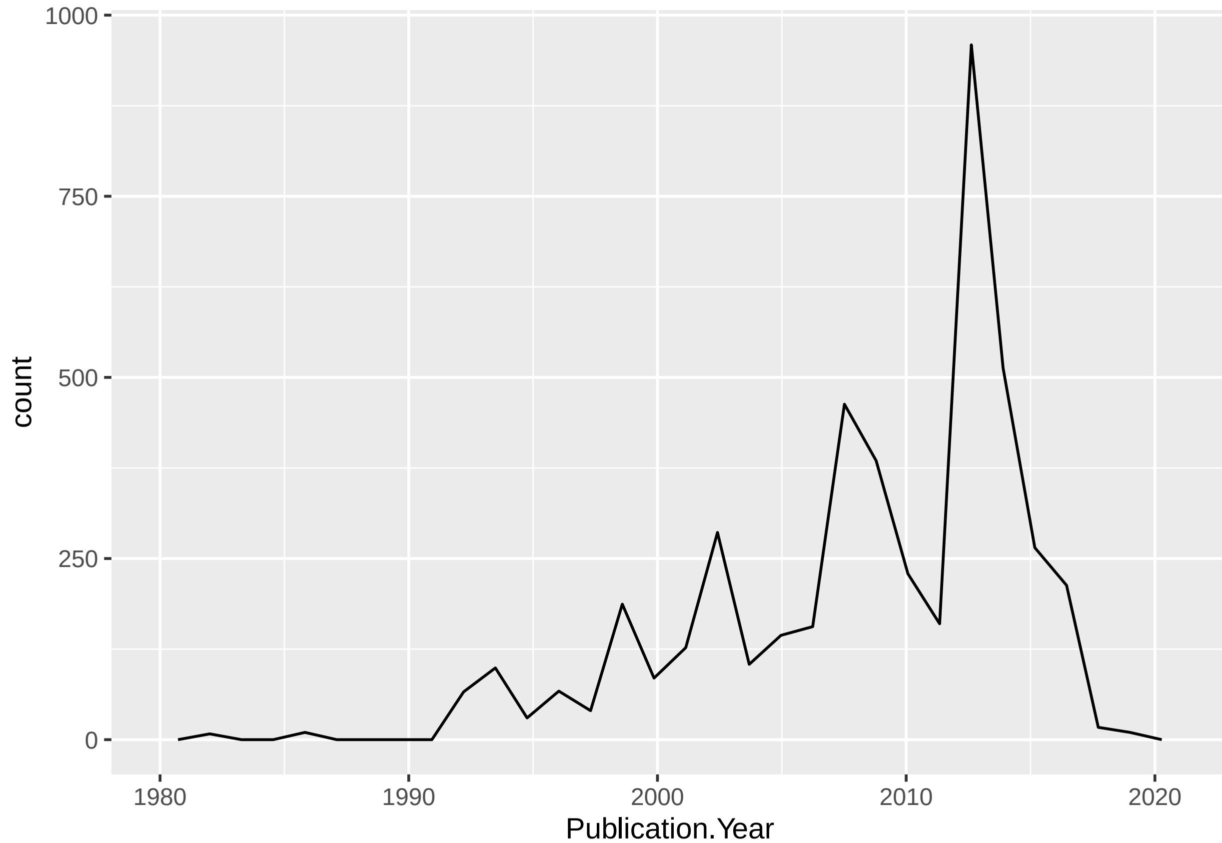
> Answer: #Conc.1..Author is categorized as a character instead of numeric because the column (which is a vector) can only store data points from one data type. Since some entries list "NR" instead of a numeric concentration, it was classified as a character.

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```r
ggplot(Neonics) +
  geom_freqpoly(aes(x = Publication.Year))
```

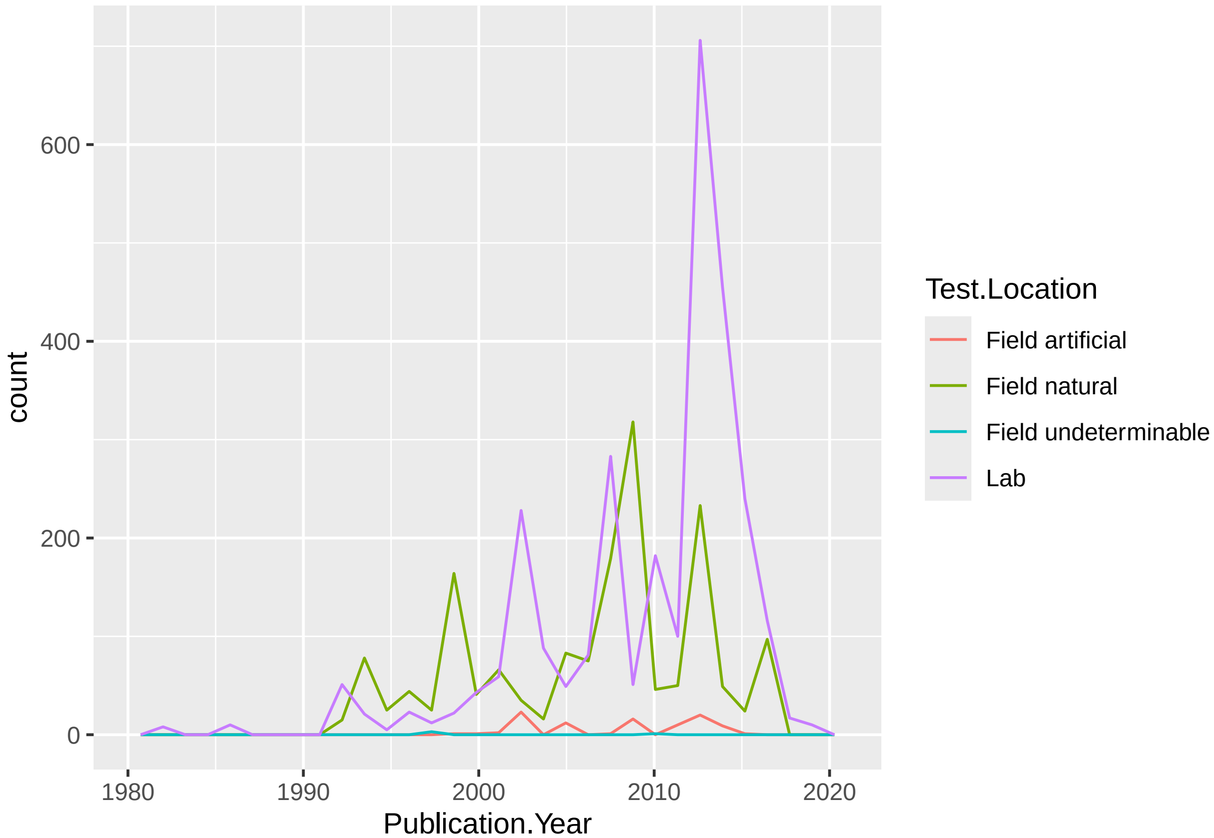
```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
ggplot(Neonics)+
  geom_freqpoly(aes(x = Publication.Year, color=Test.Location))
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

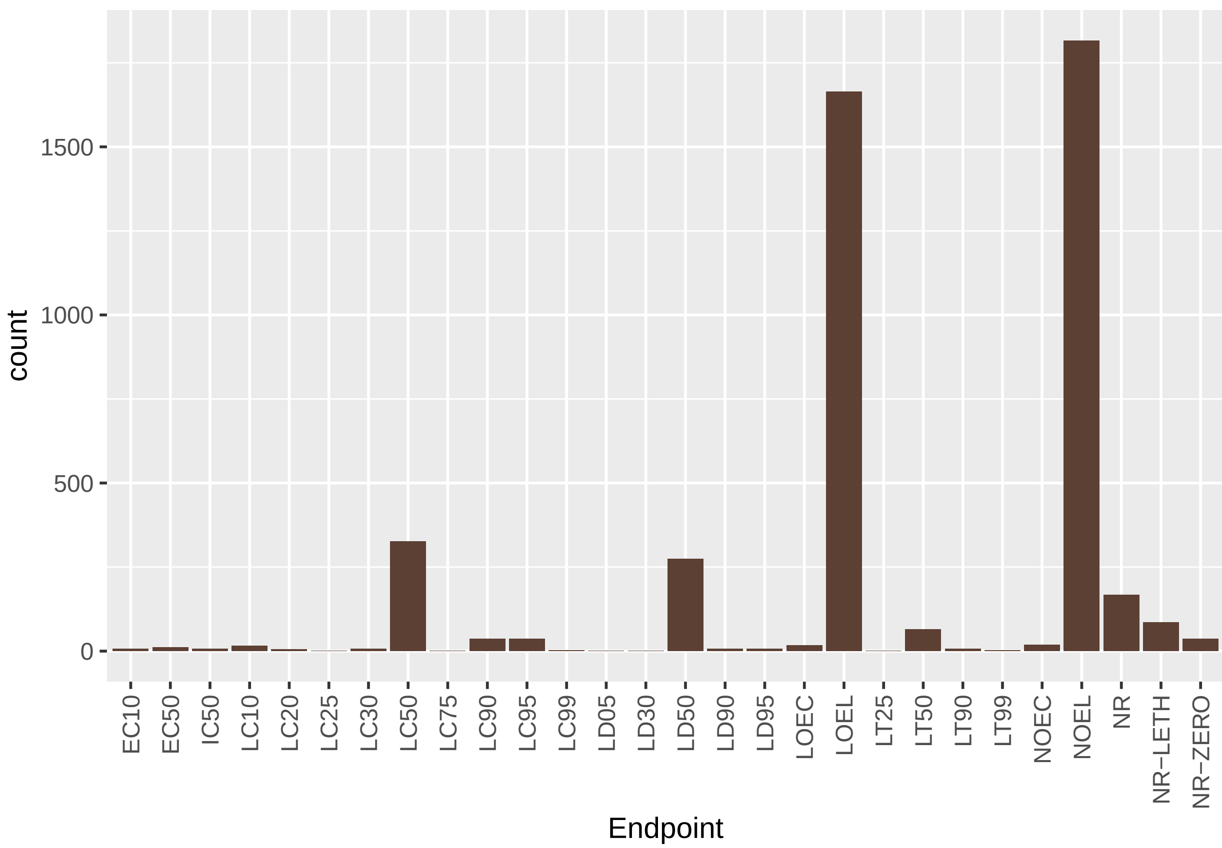


Interpret this graph. What are the most common test locations, and do they differ over time? > Answer: Field natural is the most common test location for a period right before 2010. Overall, Lab is the most common test location with a significant uptick between 2010 and 2020.

11. Create a bar graph of Endpoint counts.

[**TIP**: Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels…]

```
ggplot(data=Neonics, aes(x= Endpoint))+theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=
  geom_bar(fill ="#5C4033")
```



What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix (p.721) for more information. > Answer: The two most common end points are LOEL(lowest observable effect level) and NOEL(no observable effect level.) Both measure the effect relative to controls. —

Explore your data (Litter)

12. Determine the class of `collectDate`. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
class("collectDate")#shows that it is a character not a date
```

```
## [1] "character"
```

```
Litter$collectDate <-as.Date(Litter$collectDate, format='%Y-%m-%d')

class(Litter$collectDate) #confirmed it is a date now
```

```
## [1] "Date"
```

13. Using the `unique` function, list the different `plotIDs` sampled at Niwot Ridge.

```r
unique(Litter$plotID)
```

```
##  [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051
##  [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

```r
summary(Litter$plotID)
```

```
## NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061
##       20       19       18       15       14        8       16       17
## NIWO_062 NIWO_063 NIWO_064 NIWO_067
##       14       14       16       17
```

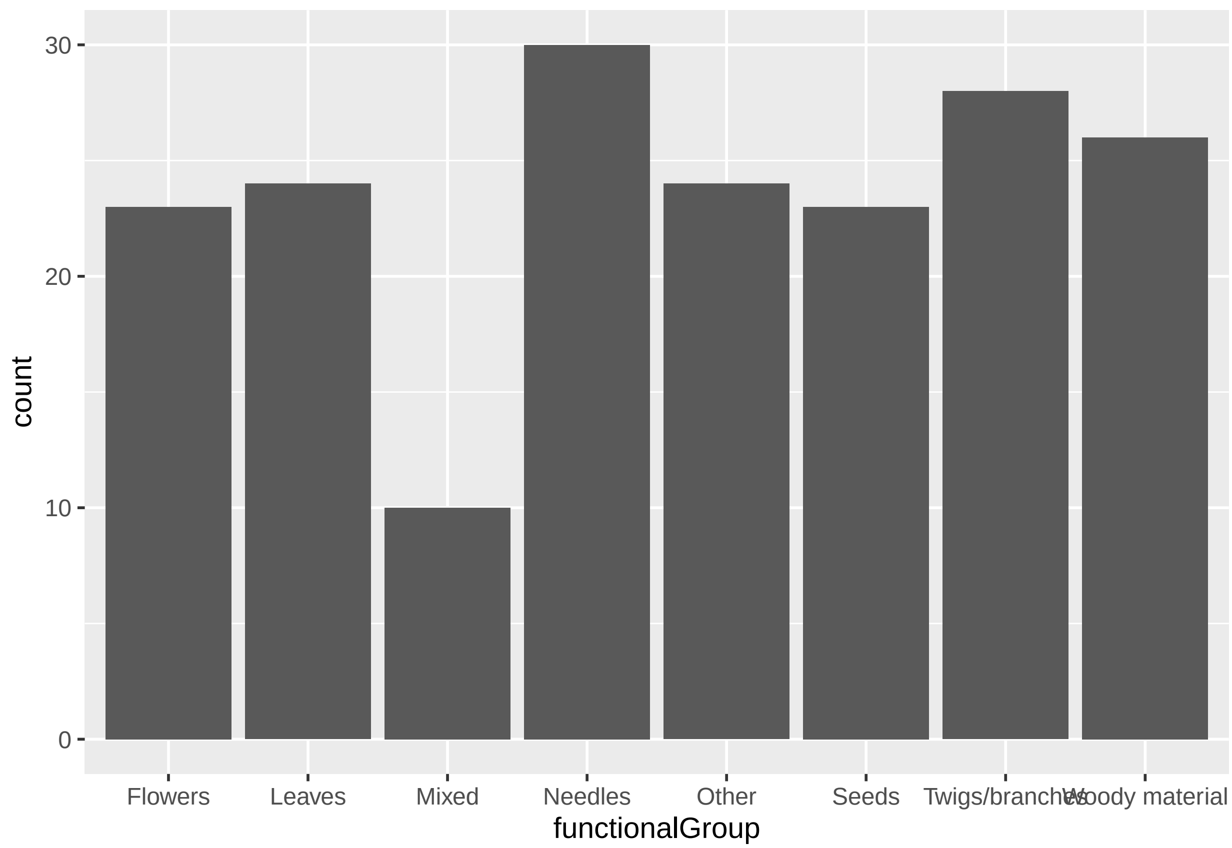
```r
class(Litter$plotID) #this is a factor
```

```
## [1] "factor"
```

How is the information obtained from `unique` different from that obtained from `summary`? > Answer:The unique function simply lists the unique IDs without duplicates. The summary function includes the number of times each unique ID appears in the data set.

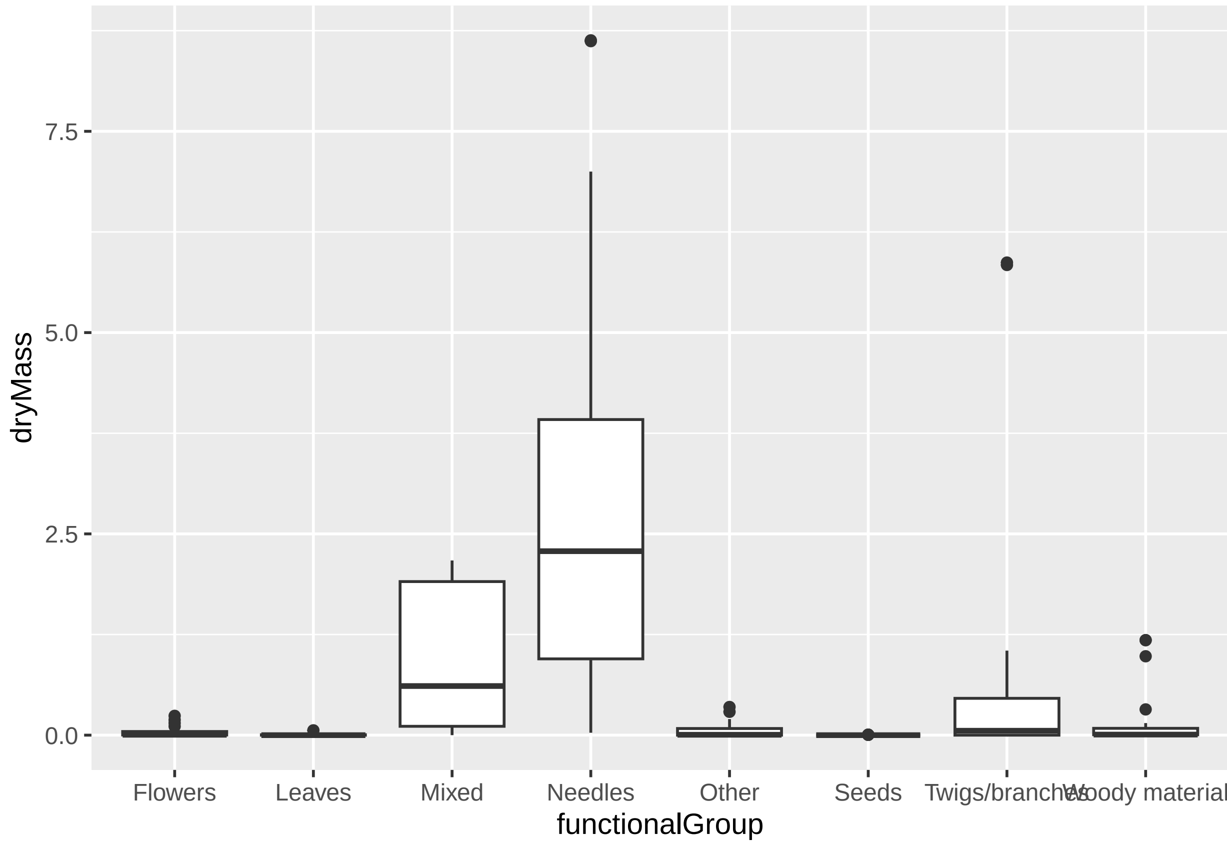
14. Create a bar graph of `functionalGroup` counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```r
ggplot(data=Litter, aes(x =functionalGroup))+
  geom_bar()
```

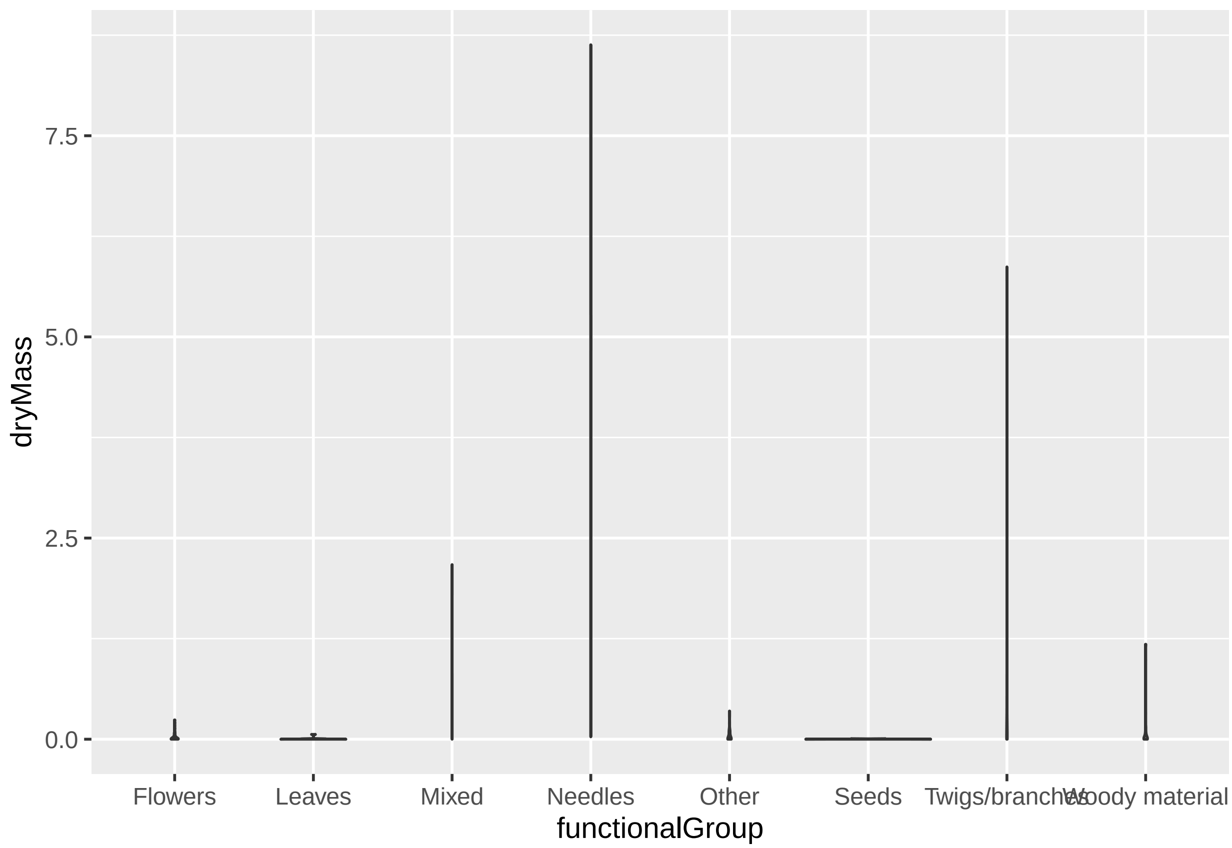


15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

```
ggplot(Litter)+
  geom_boxplot(aes(x = functionalGroup, y = dryMass))
```




```r
ggplot(Litter)+
  geom_violin(aes(x=functionalGroup, y=dryMass))
```



Why is the boxplot a more effective visualization option than the violin plot in this case? > Answer:The box plot shows the actual variation between the data points, whereas the violin plot oversimplifies to a straight line.

What type(s) of litter tend to have the highest biomass at these sites? > Answer: Needles have the highest biomass, with a median biomass of almost 2.5.